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Hard Negative Mining for Knowledge Graph Link Prediction

A Controlled Study on FB15k-237

Knowledge Graphs and Deep Learning Courses 2026

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Problem

A **knowledge graph** (KG) is a network of facts stored as triples **(subject, relation, object)**. **Link prediction** asks: given $(h, r, ?)$, which entity t is most likely to complete the triple?

Models learn by comparing correct triples against *negatives* (fabricated wrong triples). The quality of negatives is critical.

Random negatives (standard)
Obama → born_in → *Ice cream*
Obama → born_in → *Mt. Everest*
Obama → born_in → *Pacific Ocean*
Too easy. Model stops improving.

Hard negatives (ours)
Obama → born_in → *Chicago*
Obama → born_in → *Kansas City*
Obama → born_in → *Nairobi*
Plausible but wrong. Model keeps improving.

Hard negatives are wrong answers the model finds plausible, they sit just below the correct answer (Honolulu) in the ranking.

Motivation

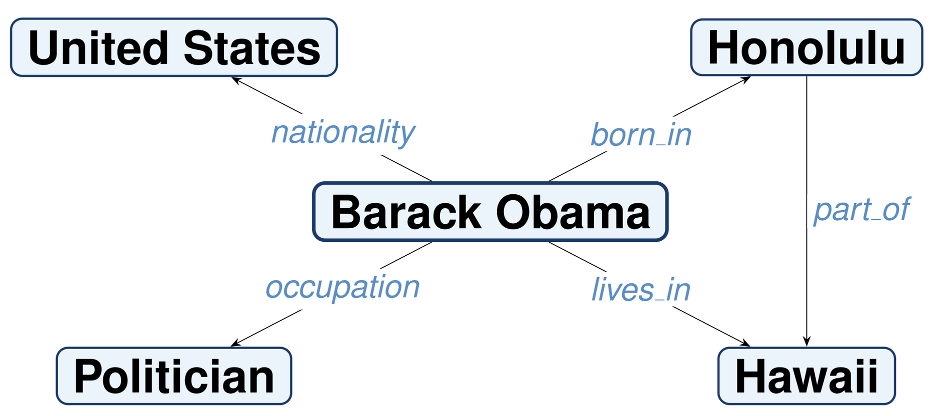
Random negatives are too easy: the model rejects them quickly and stops improving.

Hard negatives carry a stronger signal, but always picking the same plausible-but-wrong cases could over-focus training.

Question: would a **mix** of random and hard combine both effects, and at what proportion?

Dataset

FB15k-237 [2] is a widely used benchmark knowledge graph derived from Freebase, covering people, places, films, and organisations. Each node is an **entity**; each labelled edge is a **fact**.



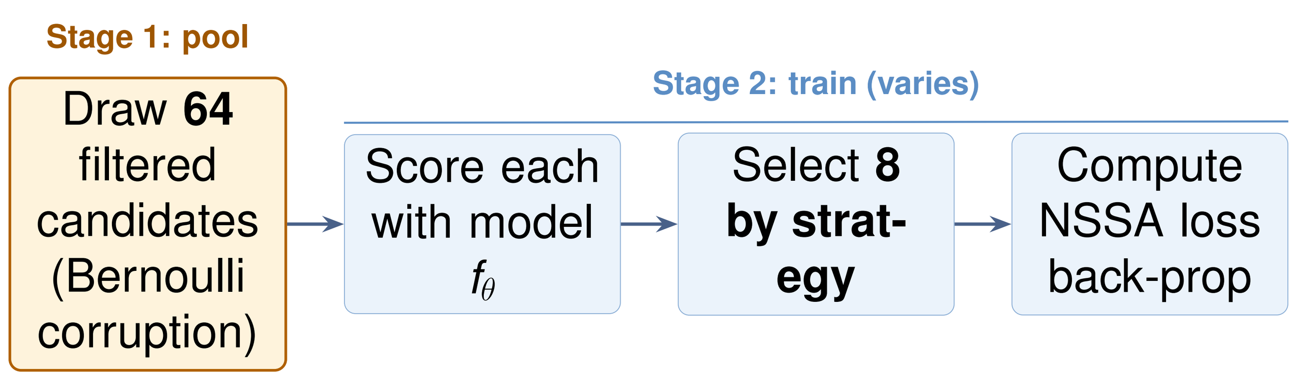
Entities:	14,505	Train triples:	272,115
Relation types:	237	Val / Test:	17,526 / 20,438

Relation frequencies range over **three orders of magnitude** (37 to 15,989 occurrences), enabling stratified evaluation by rarity.

Methodology

Controlled ablation. We hold every hyper-parameter fixed (data, model, optimiser, seed) across five training runs and **vary only the negative-selection rule**.

1. The training pipeline. For each positive triple:



2. The five strategies (only the **"Select 8 by strategy"** box differs):

Random (control)
Pick 8 of the 64 candidates *uniformly at random.*
Easy negatives, weak gradient.

Hard
Pick the 8 *highest-scoring* wrong candidates (model finds them most plausible).
Strong signal, can over-focus.

Mixed p%
p% hard + $(1-p)$% random, with $p \in \{30, 50, 70\}$.
Trade-off: accuracy + stability.

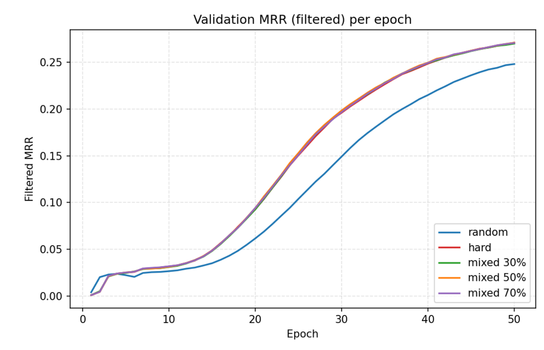
3. Setup. All strategies share the same Stage-1 pool (*zero extra compute*). RotatE [1], dim 128, batch 1024, Adam lr$=10^{-3}$, 50 epochs (patience 10), seed 42.

Results

+10.4 %
accuracy gain (MRR) with Hard negatives vs. standard random sampling

Strategy	MRR	H@1	H@3	H@10
Random (baseline)	0.270	0.189	0.300	0.428
Hard	0.298	**0.217**	0.327	0.459
Mixed 30%	0.297	0.215	0.326	0.459
Mixed 50%	0.298	0.216	**0.328**	0.460
Mixed 70%	**0.299**	**0.217**	**0.328**	**0.461**
Δ Hard − Random	+0.028	+0.028	+0.026	+0.031

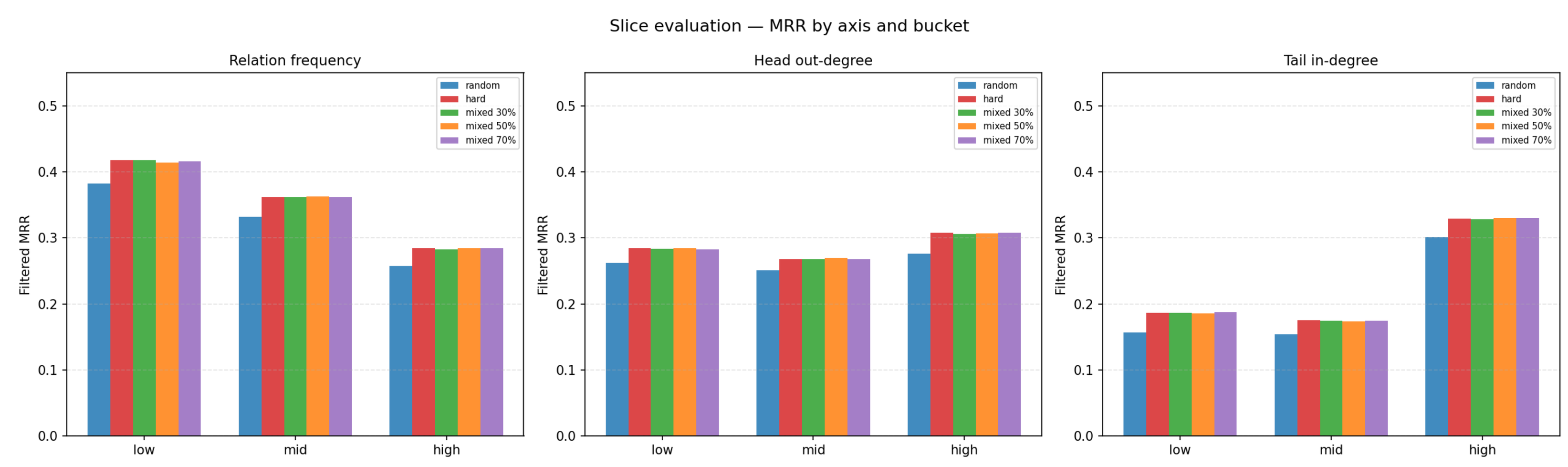
Mixed 30% hard recovers 95% of the gain at zero extra cost.



Hard / mixed strategies pull ahead after epoch 15; no instability.

Where do the gains come from?

Average MRR can hide *where* a method really helps. We slice the test set along **three axes**, relation frequency, head out-degree, tail in-degree, at the 33rd / 67th percentiles of the training distribution, and recompute MRR per tertile.



Hard and mixed variants **uniformly dominate Random in every bucket**.

Biggest gains (Hard vs. Random, in pp = percentage points): rare relations **+3.6**; high head out-degree **+3.1**; low tail in-degree **+3.0**.

Take-home: hard negatives shine on *every* slice; the single biggest gain is on rare relations (+3.6 pp), whose embeddings are the most under-trained.

Qualitative check

Tail prediction for `(Film, release_region, ?)` (filtered rank of the **gold entity**, lower is better):

Random	rank 9	Mixed 30 %	rank **1**
Hard	rank **1**	Mixed 50/70 %	rank 3

Random buries the gold answer at rank 9; Hard (and even 30 % hard) pulls it to the top.

Conclusions

- **Hard negatives are a free upgrade:** same pipeline, smarter selection step, +10.4% MRR.
- **You don't need 100% hard:** Mixed 30% (2 of 8 hard) recovers 95% of the gain with added stability.
- **Gains are largest on data-sparse slices (rare relations +3.6 pp, low tail in-degree +3.0 pp):** sparse training data leaves embeddings under-trained, so each hard-negative step is more informative.
- **Head prediction stays ∼2× harder (∼0.18 vs. ∼0.39 MRR):** FB15k-237 is dominated by **N-to-1** relations (e.g. `/film/film/genre`), so many heads compete for the same tail. This is a dataset bias no sampling rule fixes.

References

[1] Z. Sun, Z.-H. Deng, J.-Y. Nie, J. Tang. *RotatE: Knowledge Graph Embedding by Relational Rotation in Complex Space.* ICLR 2019. (model)

[2] K. Toutanova, D. Chen. *Observed versus latent features for knowledge base and text inference.* ACL Workshop 2015. (FB15k-237)

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